Assessment -3

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1) Intelligent Agents in a Smart home Automation System

Introduction:

A Smart home automation System uses Artificial Intelligence (AI) to Control lighting, temperature, and security based on users behaviours and environmental conditions. It receives real-time information from motion Sensors, temperature Sensors, and Security devices.

The four main types of intelligent agents are

1. Simple Reflex Agent
2. Model - Based Reflex Agent
3. Goal Based Agent
4. utility Based Agent.

1) Simple Reflex Agent :-

Acts only on the Current Percept using condition-action rules. for example, it. Switches on a light when a motion Sensors detects movement. It is fast but does not remember past events.

2. Model - Based Reflex Agent :-

Maintains an internal state of the environment including information that may not be directly observable For example, it remembers that a door was locked even when the lock sensor is not being Checked Continuosly

open paths, and visited junction. After each move, it updates this information using new percepts. This helps it avoids loops and choose actions based on what it has already discovered

Effective Combination :-

A model - based agent working with ves is a strong choice when the aim is to find the least - cost escape route. The agent updates its map as new information becomes available, and ves compares routes by their maze is not know in advance.

Conclusion :-

The model - based agent provides memory of the maze while the search strategy provides a systematic way to select a route, together they support safe exploration and a cost effective path to the exit.

2) Unknown Maze Navigation Adearch dtrategies

An autonomous agent navigating an cinknown maze must discover path avoid obstacles, remberber visited junctions. and find dome areas.

Cuinformed dearch dtrategies:

• Uniform Cost dearch (UCS):

Expands the node with the lowest accumulated path Cost, g(n) when step Costs are non-negative, it finds the least Cost path. It is useful when different routes have different Costs, but it may use Considerable time and memory because it explores many paths.

• Dtoratice Deeping dearch (IDS)

Runs depth-limited dearch repeatedly incresing the depth limit each time. It Combines the low memory use of depth-first dearch with the Completeness of breadth.

Agent dtate and Rational Decisions:

A model-based agent is Suitable because it maintains an internal map of the maze, recording known walls.

3) Goal - Based Agent :-

Chooses actions that help achieve a specific goal for example, it adjusts heating (or) cooling unit the perfect temperature of 22°c

4) utility - Based Agent :-

Compares possible outcomes and selects the one with the greatest overall benefit.

Best App roach : Hybrid Intelligent Agent

A hybrid agent combining reflex model - based, goal based and utility based features is the most suitable choice Reflex rules can respond immediately to danger, such as activating an alarm when a break in detected.

Conclusion :

A hybird approach is effective because it Combines quick reactions with memomory and carefull decision-making Tools makes the smart home safer, more Comfortable.